

BASEL AHMAD SHABAB

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RESEARCH PROFILE

Ph.D. student in Mining and Minerals Engineering conducting experimental research on wellbore cement integrity for underground hydrogen storage. Experience spans materials characterization, geomechanics, petrophysical testing, Python-based data analysis, and techno-economic evaluation of geologic CO₂ storage, with a focus on subsurface energy systems and applied research.

EDUCATION

Ph.D. in Mining and Minerals Engineering

Spring 2024 – Present

Virginia Tech, Blacksburg, USA (Expected Graduation: Fall 2027)

Research Area: Wellbore integrity assessment of wellbore cement for underground hydrogen storage systems.

Relevant Coursework: Numerical Modeling of Petroleum Reservoirs, Multicomponent Thermodynamics, Microfluidics and Nanofluidics.

B.S. in Petroleum and Mining Engineering

Spring 2016 – Summer 2021

Chittagong University of Engineering & Technology (CUET), Chittagong, Bangladesh

RESEARCH EXPERIENCE

Graduate Research Assistant

Spring 2024 – Present

Department of Mining & Minerals Engineering, Virginia Tech

- Characterized Class H cement (SEM-EDS, XRD, XRF, porosity/permeability, and strength) to evaluate wellbore integrity for underground hydrogen storage.
- Quantified fracture behavior of cemented paste backfill via S- and P-wave velocity measurements, generating data used to improve mine-design decisions.
- Executed all testing under strict laboratory safety and regulatory standards with zero compliance incidents.

Virtual Intern – Techno-Economic Analyst

Fall 2024

Society of Exploration Geophysicists (SEG) EVOLVE Program

- Led the techno-economic analysis for a site-scale offshore CO₂ storage project (Matagorda Island, Texas) as part of a 6-member cross-departmental team, evaluating commercial viability of storing ~15 Mt CO₂ over a 15-year injection period.
- Developed the project economic model, risk analysis, and measurement, monitoring, and verification (MMV) framework to assess long-term storage cost and project viability under current CCS tax-credit incentives.
- Compared pipeline and barge transportation scenarios and identified pipeline routing from Corpus Christi and Victoria emission sources as the preferred source-to-sink pathway.

SELECTED RESEARCH PROJECTS

Backfill Fracturing Tool for Estimating In-Situ Strength of Cemented Paste Backfill

Spring 2025

- Characterized cemented paste backfill materials by measuring Young's modulus, Poisson's ratio, and tensile strength to support development of a tool for estimating true in-situ backfill strength.

PUBLICATIONS

- Shabab, B. A., & Pandey, R. (2026). *Experimental evaluation of wellbore cement integrity in underground hydrogen storage environments*. International Journal of Hydrogen Energy. [DOI]: <https://doi.org/10.1016/j.ijhydene.2026.153957>
- Frimpong, J. A., Shabab, B. A., et al. (2025). *Fracture initiation pressure as a measure of cemented paste backfill strength*. Mining, Metallurgy & Exploration. [DOI]: <https://doi.org/10.1007/s42461-025-01257-6>
- Shabab, B. A., & Das, H. S. (2022). *A comprehensive study of low resistivity reservoirs*. 4th International Conference on Energy and Power, Dhaka, Bangladesh. [[Link](#)]

CONFERENCE PRESENTATIONS

- Shabab, B. A., & Pandey, R. (2026, June 21). *Multiscale Assessment of Wellbore Cement Integrity in Underground Hydrogen Storage* [Oral presentation]. 60th U.S. Rock Mechanics/Geomechanics Symposium (ARMA), Tucson, AZ, USA.
- Mendez, J.*, Andrade, I., Effiong, A. J., Irumhe, E. P., Lawson, B. O.*, Shabab, B. A., & Pollyea, R. (2025, March). *Site-Scale Evaluation of Geologic Carbon Storage at the Matagorda Island Leasing Area, Offshore Texas* [Poster presentation]. Carbon Capture, Utilization, and Storage (CCUS) Conference, Houston, TX, USA. (*presenting authors)

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python (Pandas, NumPy, Matplotlib), OriginPro, ImageJ/FIJI, Dragonfly, Profex, Highscore Plus
- **Geomechanics & Reservoir Tools:** Rocscience RS2, PIPESIM, PROSPER, Kingdom Petrophysics
- **Materials & Microstructural Characterization:** SEM-EDS, XRD, XRF, TGA
- **Geomechanics & Petrophysics:** Brazilian Tensile Strength Testing, Helium Pycnometry, Gas Permeability Testing, Ultrasonic P- and S-wave measurements
- **Laboratory Safety:** Compressed Gas Cylinder, Hazard Assessment, Lock-out/Tag-out, Respiratory Protection

HONORS & AWARDS

- Bishop Family Pylon Graduate Fellowship - Virginia Tech | Fall 2026 - Spring 2027
- Society of Explosive Engineers Education Foundation Scholarship | Fall 2026 - Spring 2027
- Pratt Fellowship - Virginia Tech | Fall 2024 - Spring 2025
- Academic Merit Scholarship - CUET | Top 10% ranking for 7 semesters

LEADERSHIP AND PROFESSIONAL SERVICE

Office Assistant & Challenge Course Facilitator	Summer 2026
<i>Venture Out (Recreational Sports), Virginia Tech</i>	
● Facilitated team-building and challenge-course activities for diverse participant groups while applying departmental safety and risk-management procedures. ● Inspected and documented equipment, briefed participants on activity risks and procedures, and monitored participant well-being throughout programming.	
Logistics Secretary – Association for Bangladeshi Students at Virginia Tech	Summer 2025 – Spring 2026
Vice President - CUET SPE Student Chapter	Spring 2019 – Summer 2020

CERTIFICATIONS

- American Red Cross CPR/AED for Professional Rescuers | Issued June 2026, valid through June 2028

PROFESSIONAL AFFILIATIONS

- Society of Petroleum Engineers (SPE), Society for Mining, Metallurgy & Exploration (SME)
- Spring 2024 – Present**